

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 3.2: Probability I — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.2: Probability I
Driving Question	DRIVING QUESTION: "How can Kamau — and other Kenyans facing uncertain everyday outcomes — use the mathematics of probability to make smarter, fairer, and safer decisions?"

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring the Phenomenon & Experimental Probability: Why Can't Kamau Predict His Full Bus?	Kamau's 20-trip matatu tally notebook showing passenger counts (6, 9, 14, 11, 14, 8, 14, 12, 7, 14, 10, 14, 13, 6, 14, 9, 14, 11, 8, 14) with no single rule that perfectly predicts every trip; our own coin-toss experiment showing that 30 tosses of a 10-shilling coin produced different numbers of heads and tails for each pair, with no pair getting exactly 15 of each, yet the class pooled total was close to 0.5.	Experimental probability (relative frequency) = number of times an event occurs ÷ total number of trials. Individual experiments vary — Kamau's 20 trips gave $P(\text{Full Bus}) = 8/20 = 0.4$, and each pair's coin experiment gave a slightly different $P(\text{Heads})$. However, when we pooled all pairs' results, the class total moved closer to 0.5. This shows that MORE trials give a more reliable estimate of probability, even though no single trial can be predicted with certainty.	Kamau cannot predict whether a single seat will be filled because each passenger's decision to board is an independent, uncertain event — just like each coin toss is independent. However, over many trips (or many tosses), the FRACTION of times an event happens settles toward a stable value. This stable fraction IS the experimental probability, and it gives Kamau — and us — a mathematical tool to plan and make decisions despite uncertainty, even when we cannot predict any single outcome.
Lesson 2 The Probability Scale — Placing Kamau's World on a Number Line from 0 to 1	We sorted 12 Kenyan real-life events — from a matatu carrying 20 passengers in a 14-seater (impossible) to the sun rising tomorrow in Nairobi (certain) — onto a number line from 0 to 1. We also placed Kamau's calculated probabilities ($P(\text{Full Bus}) = 0.4$, $P(\text{Not Full}) = 0.6$, $P(\text{Fewer than 8}) = 0.15$) on the same line.	Every probability must satisfy $0 \leq P(E) \leq 1$. An impossible event has $P(E) = 0$ and a certain event has $P(E) = 1$. All of Kamau's frequency ratios from his 20-trip notebook fall inside this range — none can be negative or greater than 1 — because the count of an outcome can never be less than 0 or more than the total number of trials.	Kamau cannot predict with certainty whether the next seat will be filled because the probability of a full bus is 0.4 — it sits between 0 and 1, meaning it is neither impossible nor guaranteed. The probability scale shows mathematically that Kamau's uncertainty is real and bounded: he will never face an outcome with $P < 0$ or $P > 1$, so even though he cannot be 100% sure, he can say precisely how sure he is.
Lesson 3	Learners constructed a 2x2 two-way table	The sample space (probability space) is the	Kamau cannot predict with certainty

Probability Space (Sample Space): Listing Every Possibility for Kamau's Route	for Kamau's two-stop (Mlolongo and Athi River) model, filling in all four ordered outcome pairs (TT, TE, ET, EE), and built a 6×6 grid listing all 36 ordered pairs for two dice at a school game stall fundi fair.	complete set of all possible outcomes of a random experiment; theoretical probability of an event equals the number of favourable outcomes divided by the total number of outcomes in the sample space; a two-way table guarantees exhaustive, non-repetitive enumeration of outcomes for two-stage experiments.	whether a seat will be filled at any given stop because all four outcomes in his sample space {TT, TE, ET, EE} are genuinely possible; however, the sample space gives him a mathematical tool to assign theoretical probabilities to each outcome, which approximate — but do not exactly match — the experimental frequencies recorded in his tally notebook.
Lesson 4 Theoretical Probability and Single Events: What Should Kamau Expect?	Learners computed theoretical probabilities from Lesson 3 sample space grids ($P(\text{sum of two dice} = 7) = 6/36$) and extracted experimental relative frequencies from Kamau's 20-trip tally notebook, recording both values side-by-side in a Convergence Worksheet. They also observed the teacher-facilitated public examination of the 10/14 vs 14/20 misconception on the Convergence Chart.	Learners learned that theoretical probability is defined as $P(E) = (\text{number of favourable outcomes}) \div (\text{total number of equally likely outcomes})$; that theoretical probability values range between 0 and 1; that Kamau's theoretical probability of a randomly chosen seat being occupied is $10/14 \approx 0.71$; and that as the number of experimental trials increases, experimental probability converges toward theoretical probability (Law of Large Numbers, stated informally).	Learners explained in writing and discussion why Kamau's 20-trip experimental frequency does not exactly match the theoretical probability 10/14, connecting this to the insufficiency of small sample sizes; why 10/14 and 14/20 answer fundamentally different questions about Kamau's scenario; and why the sample space grid from Lesson 3 is the tool that makes theoretical probability countable and exact. They updated their DQB with the consensus question: 'Are theoretical and experimental probabilities ever exactly the same — and if so, when?'
Lesson 5 Mutually Exclusive Events and the Addition Law	Learners investigated a Kisumu Day School prize-draw sample space and Kamau's 20-trip matatu tally to identify events that share no outcomes.	Two events are mutually exclusive when they cannot occur at the same time in a single trial (they share no sample space outcomes), and the probability of either occurring is found using the Addition Law: $P(A \text{ or } B) = P(A) + P(B)$.	Learners explained why $P(\text{fully booked OR fewer than 7 passengers}) = P(F) + P(L) = 8/20 + 2/20 = 1/2$ for Kamau's matatu, and why the formula works only because a trip cannot be both fully booked and nearly empty at the same time.
Lesson 6 Independent Events and the Multiplication Law: Can Kamau Count on Both Trips?	Learners conducted a paired coin-toss experiment (20 trials per pair, ~160 trials pooled class-wide) using Kenyan 5-shilling coins, recording outcomes in a four-row tally table and building a 2×2 sample space grid. They observed that the frequency of each of the four ordered outcomes (HH, HT, TH, TT) was approximately equal ($\sim 1/4$ each), and that the conditional frequency of Heads on the second toss given Heads on the first toss was approximately equal to the unconditional frequency of Heads — providing empirical evidence that the two tosses are independent.	Learners formalised the definition of independent events — two events are independent when the occurrence of one does not alter the probability of the other — and derived the multiplication law: $P(A \text{ and } B) = P(A) \times P(B)$. They applied this law in two Kenyan contexts: (i) Amina the Kericho tea farmer calculating $P(\text{rain on both Monday and Tuesday}) = 3/5 \times 3/5 = 9/25 = 0.36$, supported by a fully labelled two-stage tree diagram; and (ii) Kamau's matatu route, calculating $P(\text{full morning AND full afternoon}) = 2/5 \times 2/5 = 4/25 = 0.16$, interpreted as 'roughly once a working week'. Learners also previewed dependent events through the queue-without-	Learners explained, through their cumulative model page and DQB claim cards, that two events are independent when knowing the result of one gives no new information about the probability of the other, and that this independence allows probabilities to be combined simply by multiplication. They articulated — in plain language — what the probability of 0.16 means for Kamau's operational planning, and distinguished this independent scenario from the dependent queue-selection preview. The DQB was updated with the new claim: 'Two events are independent when the probability of the second is the same whether or not the first occurred —

		replacement contrast, distinguishing $P(\text{second} = \text{woman} \mid \text{first} = \text{woman}) = 4/13 \neq 5/14$ from the independent case.	and then $P(A \text{ and } B) = P(A) \times P(B)$ gives us the exact combined chance.'
Lesson 7 Probability Tree Diagrams: Mapping All of Kamau's Possible Outcomes	Learners observed a two-stage probability tree diagram built step-by-step for Kamau's matatu boarding scenario (Stop 1: Mlolongo; Stop 2: Athi River), with $P(\text{Filled}) = 7/10$ and $P(\text{Empty}) = 3/10$ at each stop. They watched all four terminal branch probabilities computed as products and verified that the four values sum to 1. They also observed a second tree diagram for a Nairobi CBD road-safety scenario showing $P(\text{undesignated crossing AND near-miss}) = 6/25$.	Learners learned that: (1) a tree diagram displays ALL possible outcomes of sequential events simultaneously; (2) probabilities are MULTIPLIED along each branch path (multiplication rule for independent events); (3) probabilities are ADDED across branches to find the probability of any condition spanning multiple outcomes; (4) the sum-to-1 check is a built-in error detector; (5) $P(\text{both seats filled across two stops}) = 49/100$; $P(\text{at least one seat filled}) = 91/100$; $P(\text{exactly one seat filled}) = 42/100$; (6) crossing at an undesignated point in Nairobi is 6× more likely to result in a near-miss than crossing at a designated point.	Learners explained that tree diagrams give Kamau a complete mathematical picture of all two-stop boarding combinations, replacing guesswork with exact probabilities. The multiplication rule works because independent events combine by scaling — each successive probability shrinks the likelihood of a specific sequence. The addition rule works because mutually exclusive terminal outcomes can be combined without overlap. The road-safety tree shows that probability is not just an academic tool — it quantifies real civic risk and can inform policy, safety campaigns, and personal decisions about where to cross the road in Nairobi.
Lesson 8 Application, Games, and Unit Synthesis — Advising Kamau with the Full Power of Probability	Students observed that Kamau's 20-trip dataset yields an experimental probability of $8/20 = 0.4$ for a fully booked trip, and that this value is dramatically different from what a naive independent-seats model predicts (≈ 0.009) — revealing that real passenger boarding is clustered, not independent. Students also observed, through peer game evaluation, that a game is only mathematically fair when every player's theoretical probability of winning is equal, and that this must be proven with a calculation, not assumed.	Learners consolidated the complete probability framework: experimental probability estimates likelihood from data and converges toward theoretical probability as trials increase (Law of Large Numbers); theoretical probability assumes equally likely outcomes and a known sample space; the Addition Law handles mutually exclusive events; the Multiplication Law and tree diagrams handle independent sequential events; and when a theoretical model's prediction contradicts experimental data, the model's assumptions (here, independence) must be questioned. A Probability Report translates all of this into actionable advice for a decision-maker like Kamau.	The mathematics of probability explains why Kamau cannot predict any single trip with certainty — each trip is a random event influenced by factors (school terms, market days, social groups) that create clustering rather than independence — yet it empowers him to plan wisely: with $P(\text{full trip}) \approx 0.4$, he can expect roughly 2 fully booked trips out of every 5 morning departures, budget for a second vehicle on approximately 8 days per school-term month, and communicate confidently to his employer that uncertainty is not ignorance but a quantifiable, manageable feature of his route.

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